Start here | sjf_non.c

```c
#include<stdio.h>

int main()
{
    int n,i,j,time=0,small;
    int at[20],bt[20],ct[20],tat[20],wt[20],done[20]={0};

    printf("Enter number of processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        printf("Enter AT and BT for P%d: ",i+1);
        scanf("%d%d",&at[i],&bt[i]);
    }

    for(i=0;i<n;i++)
    {
        small=-1;

        for(j=0;j<n;j++)
        {
            if(done[j]==0 && at[j]<=time)
            {
                if(small==-1 || bt[j]<bt[small])
                    small=j;
            }
        }

        if(small==-1)
        {
            time++;
            i--;
            continue;
        }

        time+=bt[small];
```

Start here | sjf_non.c

```c
    for(i=0;i<n;i++)
    {
        small=-1;

        for(j=0;j<n;j++)
        {
            if(done[j]==0 && at[j]<=time)
            {
                if(small==-1 || bt[j]<bt[small])
                    small=j;
            }
        }

        if(small==-1)
        {
            time++;
            i--;
            continue;
        }

        time+=bt[small];
        ct[small]=time;
        tat[small]=ct[small]-at[small];
        wt[small]=tat[small]-bt[small];

        done[small]=1;
    }

    printf("\nP\tAT\tBT\tCT\tTAT\tWT\n");

    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",i+1,at[i],bt[i],ct[i],tat[i],wt[i]);

    return 0;
}
```

C:\Users\BMSCECSE-L3-18\Do

```
Enter number of processes: 4
Enter AT and BT for P1: 0
7
Enter AT and BT for P2: 0
3
Enter AT and BT for P3: 0
4
Enter AT and BT for P4: 0
6

P       AT      BT      CT      TAT     WT
P1      0       7       20      20      13
P2      0       3       3       3       0
P3      0       4       7       7       3
P4      0       6       13      13      7

Process returned 0 (0x0)   execution time : 56.441 s
Press any key to continue.
```

Start here | sjf_pre.c

```c
#include<stdio.h>

int main()
{
    int n,i,time=0,remain,small;
    int at[20],bt[20],rt[20],ct[20],tat[20],wt[20];

    printf("Enter number of processes: ");
    scanf("%d",&n);

    remain=n;

    for(i=0;i<n;i++)
    {
        printf("Enter AT and BT for P%d: ",i+1);
        scanf("%d%d",&at[i],&bt[i]);
        rt[i]=bt[i];
    }

    while(remain!=0)
    {
        small=-1;

        for(i=0;i<n;i++)
        {
            if(at[i]<=time && rt[i]>0)
            {
                if(small==-1 || rt[i]<rt[small])
                    small=i;
            }
        }

        if(small==-1)
        {
            time++;
            continue;
```

Start here | *sjf_pre.c

```c
        small=-1;

        for(i=0;i<n;i++)
        {
            if(at[i]<=time && rt[i]>0)
            {
                if(small==-1 || rt[i]<rt[small])
                    small=i;
            }
        }

        if(small==-1)
        {
            time++;
            continue;
        }

        rt[small]--;
        time++;

        if(rt[small]==0)
        {
            remain--;
            ct[small]=time;
            tat[small]=ct[small]-at[small];
            wt[small]=tat[small]-bt[small];
        }

    }
    printf("\nP\tAT\tBT\tCT\tTAT\tWT\n");

    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",i+1,at[i],bt[i],ct[i],tat[i],wt[i]);
    return 0;
}
```

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```
Enter number of processes: 4
Enter AT and BT for P1: 0
7
Enter AT and BT for P2: 0
3
Enter AT and BT for P3: 0
4
Enter AT and BT for P4: 0
6

P       AT      BT      CT      TAT     WT
P1      0       7       20      20      13
P2      0       3       3       3       0
P3      0       4       7       7       3
P4      0       6       13      13      7

Process returned 0 (0x0)   execution time : 16.436 s
Press any key to continue.
```

art here ✕ *sjf_pre.c ✕ prio_non.c ✕

```c
#include<stdio.h>

int main()
{
    int n,i,j,time=0,high;
    int at[20],bt[20],pr[20],ct[20],tat[20],wt[20],done[20]={0};

    printf("Enter number of processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        printf("Enter AT BT Priority for P%d: ",i+1);
        scanf("%d%d%d",&at[i],&bt[i],&pr[i]);
    }

    for(i=0;i<n;i++)
    {
        high=-1;

        for(j=0;j<n;j++)
        {
            if(done[j]==0 && at[j]<=time)
            {
                if(high==-1 || pr[j]<pr[high])
                    high=j;
            }
        }

        if(high==-1)
        {
            time++;
            i--;
            continue;
        }

        time+=bt[high];
```

Start here | *sjf_pre.c | prio_non.c

```c
    {
        high=-1;

        for(j=0;j<n;j++)
        {
            if(done[j]==0 && at[j]<=time)
            {
                if(high==-1 || pr[j]<pr[high])
                    high=j;
            }
        }

        if(high==-1)
        {
            time++;
            i--;
            continue;
        }

        time+=bt[high];
        ct[high]=time;
        tat[high]=ct[high]-at[high];
        wt[high]=tat[high]-bt[high];

        done[high]=1;
    }

    printf("\nP\tAT\tBT\tPR\tCT\tTAT\tWT\n");

    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1,at[i],bt[i],pr[i],ct[i],tat[i],wt[i]);

    return 0;
}
```

C:\Users\BMSCECSE-L3-18\Do

```
Enter number of processes: 4
Enter AT BT Priority for P1: 0
7
2
Enter AT BT Priority for P2: 0
3
3
Enter AT BT Priority for P3: 0
4
1
Enter AT BT Priority for P4: 0
6
4

P       AT      BT      PR      CT      TAT     WT
P1      0       7       2       11      11      4
P2      0       3       3       14      14      11
P3      0       4       1       4       4       0
P4      0       6       4       20      20      14

Process returned 0 (0x0)   execution time : 72.433 s
Press any key to continue.
```

Block Instruction

pre.c | prio_non.c | prio_pre.c

```c
#include<stdio.h>

int main()
{
    int n,i,time=0,remain,high;
    int at[20],bt[20],pr[20],rt[20],ct[20],tat[20],wt[20];
    float avgtat,avgwt,ttat=0,twt=0;

    printf("Enter number of processes: ");
    scanf("%d",&n);

    remain=n;

    for(i=0;i<n;i++)
    {
        printf("Enter AT BT Priority for P%d: ",i+1);
        scanf("%d%d%d",&at[i],&bt[i],&pr[i]);
        rt[i]=bt[i];
    }

    while(remain!=0)
    {
        high=-1;

        for(i=0;i<n;i++)
        {
            if(at[i]<=time && rt[i]>0)
            {
                if(high==-1 || pr[i]<pr[high])
                    high=i;
            }
        }

        if(high==-1)
        {
            time++;
            continue;
```

Start here | *sjf_pre.c | prio_non.c | prio_pre.c

```c
        if(high==-1)
        {
            time++;
            continue;
        }

        rt[high]--;
        time++;

        if(rt[high]==0)
        {
            remain--;
            ct[high]=time;
            tat[high]=ct[high]-at[high];
            wt[high]=tat[high]-bt[high];
        }
    }
    for(i=0;i<n;i++)
    {
        ttat=ttat+tat[i];
        twt=twt+wt[i];
    }
    avgtat=ttat/n;
    avgwt=twt/n;
    printf("\nP\tAT\tBT\tPR\tCT\tTAT\tWT\n");

    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1,at[i],bt[i],pr[i],ct[i],tat[i],wt[i]);
    printf("Avarage TAT:%f",avgtat);
    printf("Avarage WT:%f",avgwt);
    return 0;
}
```

C:\Users\BMSCECSE-L3-18\Do

```
Enter number of processes: 4
Enter AT BT Priority for P1: 0
7
2
Enter AT BT Priority for P2: 0
3
3
Enter AT BT Priority for P3: 0
4
1
Enter AT BT Priority for P4: 0
6
4

P       AT      BT      PR      CT      TAT     WT
P1      0       7       2       11      11      4
P2      0       3       3       14      14      11
P3      0       4       1       4       4       0
P4      0       6       4       20      20      14
Avarage TAT:12.250000Avarage WT:7.250000
Process returned 0 (0x0)   execution time : 15.410 s
Press any key to continue.
```